An introduction to Python language

Constructor
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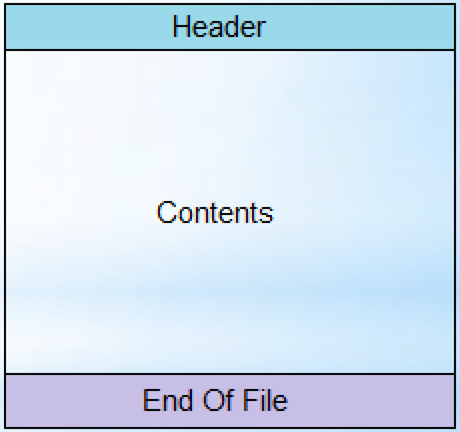
Files (few terms)

file: is a contiguous set of bytes used to store data. This data is organized in a specific format and can be anything as simple as a text file or as complicated as a program executable.

Files on most modern file systems are composed of three main parts:

- **Header:** metadata about the contents of the file (file name, size, type,..)
- **Data:** contents of the file as written by the creator or editor
- **End of file (EOF):** special character that indicates the end of the file



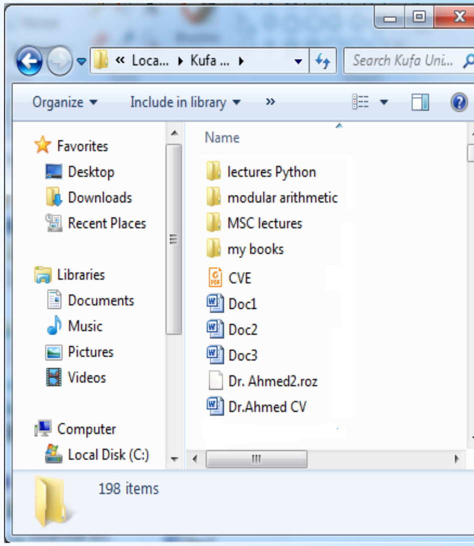
Files (few terms)

Files in python:

- **Text files**
  - Text file stores information in **ASCII OR UNICODE** character. In text file everything will be stored as a character.
  - each line is terminated by special character called **EOL**. In text file some translation takes place when this EOL character is read or written. In python EOL is '\n' or '\r' or combination of both
- **Binary files**
  - It stores the information in the same format as in the memory i.e. data is stored according to its data type so no translation occurs.
  - In binary file there is no delimiter for a new line.
  - Binary files are faster and easier for a program to read and write than text files.
  - Data in binary files cannot be directly read, it can be read only through python program.

Files (few terms)

For accessing the file we need to three major parts:
Folder Path: folder location on the file system that contain the file
File Name: the actual name of the file
File Extension: the file type its it is few characters after final period.



Examples: **Folders**

'F:\\Kufa University\\lectures Python\\lecture1.txt'

Root directory or Drive

Folder path:

File Name:

File Extension:

Notes:

- Folder is container of files or other subfolders.
- A directory is a folder
- Some file extensions are: txt, ppt, pptx, doc, docx, pdf ….

Files (few terms)

Notes:

- Folder is container of files or other subfolders.
- A directory is a folder
- Some file extensions are: txt, ppt, pptx, doc, docx, pdf …
- In python the full path can be written as:

‘F:\\Kufa University\\lectures Python\\lecture1.txt’

‘F:/Kufa University/lectures Python/lecture1.txt’

r‘F:\Kufa University\lectures Python\lecture1.txt’

Files (its location)

For dealing with folders, we need to importing os module:

- ❑ Create folder:
  os.mkdir(path)

- ❑ Delete empty folder:
  os.rmdir(path)

- ❑ Getting all files and folders in a folder (level one):
  os.listdir(path)

- ❑ Checking is file exist or not:
  os.path.isfile(path)

- ❑ Checking is folder exist or not:
  os.path. isdir(path)

Files (folders examples)

Ex1: Create folder with name “lecture” in the directory “C:\ ali” :

```
import os
os.mkdir("C:\\ali\\lecture")
```

Ex2: delete the folder “lecture” from the directory “C:\ ali” :

```
import os
os.rmdir("C:\\ali\\lecture")
```

Ex3: add folder “lecture” and inside it folder “1” in the directory “C:\ ali” :

```
import os
os.mkdir("C:\\ali\\lecture")
os.mkdir("C:\\ali\\lecture\\1")
```

Ex4: Get all files and folders in the directory “C:\ ali\lecture” in a list called L then separate the files in list F and folders in list D:

```
import os
mypath="C:\\ali\\lecture"
myfiles = os.listdir(mypath)
F=[]
D=[]
for f in myfiles:
    if os.path.isfile(mypath+"\\"+f):
        F.append(f)
    else:
        D.append(f)
```

Files (processing)

Steps in Data File Handling

- **OPENING FILE:**

  We should first open the file for read or write by specifying the name of file and mode.

- **PERFORMING READ/WRITE:**

  Once the file is opened now we can either read or write for which file is opened using various functions available

- **CLOSING FILE**

  After performing operation we must close the file and release the file for other application to use it

Files (open and close file)

For open a file use open: open(file, mode='r', buffering=-1, encoding=None, errors=None, newline=None, closefd=True, opener=None)

The important arguments are:

- file: refer to the full path of file with its name
- Mode: the mode in which the file is opened (the default is ‘r’ read):
- Encoding : encoding is the name of the encoding used to decode or encode the file. This should only be used in text mode. For Arabic use “utf-8”, “utf-16”, or 'utf-8-sig'

For closing file use:

Filename.close()

It release the file and save the any write to physical disk.

Note: open function is built-in function used standalone while close() must be called through file handle

Files (open file)

File mode

Text	Binary	Description	Notes
‘r’	‘rb’	Read only	File must exists, otherwise Python raises I/O errors
‘w’	‘wb’	Write only	If file not exists, file is created If file exists, python will truncate existing data and overwrite the file.
‘a’	‘ab’	Append	File is in write mode only, new data will be added to the end of existing data i.e. no overwriting. If file not exists it is created
‘r+’	‘r+b’ or ‘rb+’	Read and write	File must exists otherwise error is raised Both reading and writing can take place
w+	‘w+b’ or ‘wb+’	Write and read	File is created if not exists, if exists data will be truncated, both read and write allowed
‘a+’	‘a+b’ or ‘ab+’	Write and read	Same as above but previous content will be retained and both read and write.

Files (open and close file)

Examples

```
g=open("C:\\ali\\lecture\\a2.txt")

h=open("C:\\ali\\lecture\\a3.txt",'w')

t=open("C:\\ali\\lecture\\a4.txt",'r+')

f=open("C:\\ali\\lecture\\a1.txt",'r+',encoding='utf-16')
# any processing here
g.close()
h.close()
t.close()
f.close()
```

Files (Reading from File)

To read from file, python provide many functions like :

Filehandle.read([n]) : reads and return n bytes, if n is not specified it reads entire file.

Filehandle.readline([n]) : reads a line of input. If n is specified reads at most n bytes. Read bytes in the form of string ending with line character or blank string if no more bytes are left for reading

Filehandle.readlines(): reads all lines and returns them in a list

Files (Reading from File)

Example 1: read()

To read from file, python provide many functions like :

cv - Notepad

File Edit Format View Help

```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

```python
f=open("C:\\ali\\lecture\\cv.txt")
s1=f.read(11)
s2=f.read(5)
s3=f.read()
print("s1:", s1)
print("s2:",s2)
print("s3:",s3)
f.close()
```

```
s1: Dr. Ahmed H
s2: ussei
s3: n Aliwy is associate professor in University of
Kufa.
He is worked in university of technology/Baghdad fr
om 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

Files (Reading from File)

Example 2: read()

To read from file, python provide many functions like :

cv - Notepad

File Edit Format View Help

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f=open("C:\\ali\\lecture\\cv.txt")
s1=f.read()
s2=f.read(11)
s3=f.read(5)
print("s1:", s1)
print("s2:",s2)
print("s3:",s3)
f.close()
```

```
s1: Dr. Ahmed Hussein Aliwy is associate professor
in University of Kufa.
He is worked in university of technology/Baghdad fr
om 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
s2:
s3:
```

Files (Reading from File)

Example 3: readline()

To read from file, python provide many functions like :

cv - Notepad

File Edit Format View Help

```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

```python
f=open("C:\\ali\\lecture\\cv.txt")
L1=f.readline()
L2=f.readline()
L3=f.readline(10)
print(L1)
print(L2)
print(L3)
f.close()
```

```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.

He is worked in university of technology/Baghdad
from 2003 to 2015.

He got his
```

Files (Reading from File)

Example 4: readline()

To read from file, python provide many functions like :

cv - Notepad

File Edit Format View Help

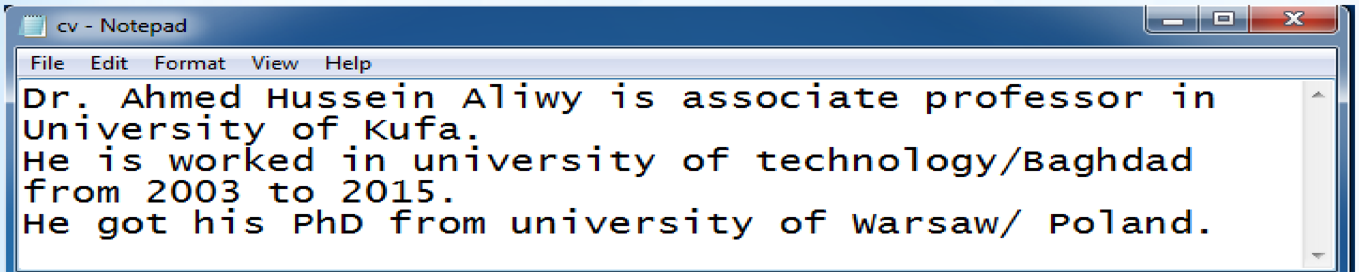
```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

```python
f=open("C:\\ali\\lecture\\cv.txt")
L1=f.readline()
L2=f.readline()
L3=f.readline(10)
print(L1,end='')
print(L2,end='')
print(L3,end='')
f.close()
```

```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his
```

Files (Reading from File)

Example 5: read files line by line


cv - Notepad

File Edit Format View Help

Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.

```python
f=open("C:\\ali\\lecture\\cv.txt")
for st in f:
    print(st,end='')
```

```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

Files (Reading from File)

Example 6: readlines()

cv - Notepad

File Edit Format View Help

```
Dr. Ahmed Hussein Aliwy is associate professor in
University of Kufa.
He is worked in university of technology/Baghdad
from 2003 to 2015.
He got his PhD from university of Warsaw/ Poland.
```

```python
f=open("C:\\ali\\lecture\\cv.txt")
content=f.readlines()
print(content)
print("first line is:",content[0])
print("second line is:",content[1])
```

```
['Dr. Ahmed Hussein Aliwy is associate professor
in University of Kufa.\n', 'He is worked in unive
rsity of technology/Baghdad from 2003 to 2015.\n'
, 'He got his PhD from university of Warsaw/ Pola
nd.']
first line is: Dr. Ahmed Hussein Aliwy is associa
te professor in University of Kufa.

second line is: He is worked in university of tec
hnology/Baghdad from 2003 to 2015.
```

Files (writing to File)

To write to file, python provide many functions like :

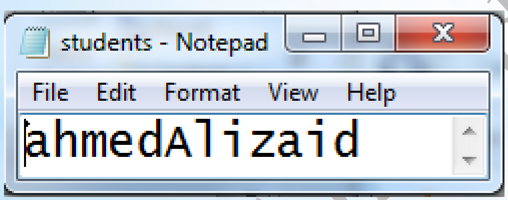
Name	Syntax	Description
write()	Filehandle.write(S1)	Writes string S1 to file referenced by filehandle
Writelines()	Filehandle.writelines(L)	Writes all string in List L as lines to file referenced by filehandle.

Files (Writing to File)

Example-1: write() using “w” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'w')
for i in range(3):
    name=input('Enter the name:   ')
    f.write(name)
f.close()
```

```
Enter the name:   ahmed
Enter the name:   Ali
Enter the name:   zaid
```

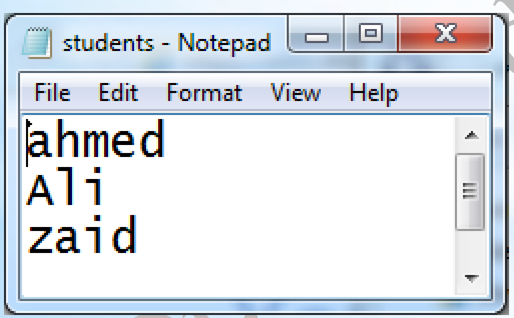


Files (Writing to File)

Example-1: write() using “w” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'w')
for i in range(3):
    name=input('Enter the name:  ')
    f.write(name)
    f.write('\n')
f.close()
```

```
Enter the name:   ahmed
Enter the name:   Ali
Enter the name:   zaid
```

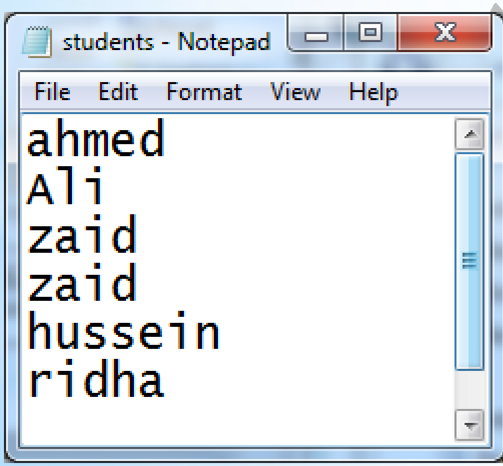


Files (Writing to File)

Example-1: write() using “a” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'a')
for i in range(3):
    name=input('Enter the name:  ')
    f.write(name)
    f.write('\n')
f.close()
```

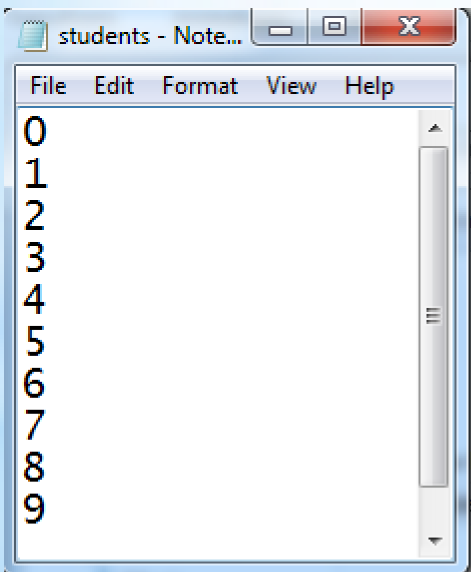
```
Enter the name:  zaid
Enter the name:  hussein
Enter the name:  ridha
```



Files (Writing to File)

Example-1: writelines () using “w” mode

```
f=open("C:\\ali\\lecture\\students.txt", 'w')
L=[]
for i in range(10):
    L.append(str(i)+'\n')
f.writelines(L)
f.close()
```



Files (exercise)

If you have a text file called “history.txt" , do the following separatly:

1- Write a function in python to read the content from a text file “history.txt" line by line and display the same on screen.

```
def read_file():
    file = open("history.txt","r")
    for line in file:
        print(line, end="")
    file.close()

read_file()
```

2- Write a function in python to count the number of lines from a text file "story.txt" which is not starting with an alphabet “A“

3- Write a function in Python to count and display the total number of words in a text file.

```
def count_words():
    file = open("history.txt","r")
    count = 0
    for line in file:
        words = line.split()
        count += len(words)
    print("Total words are",count)

count_words()
```

Files (exercise)

If you have a text file called “history.txt" , do the following separately:

4- Write a function display_words() in python to read lines from the text file , and display those words, which are less than 4 characters.

5- Write a function in Python to count the words “and" and “or" present in the text file . [Note that the words “and" and “or" are complete words]

6- Write a function in Python to count words in a text file those are ending with alphabet "f".

7- Write a function in Python to count uppercase character in a text file.

8- A text file named "matter.txt" contains some text, which needs to be displayed such that every next character is separated by a symbol "#". Write a function definition for hash_display() in Python that would display the entire content of the file in the desired format.

Example :
If the file history.txt has the following content stored in it :
THE WORLD IS ROUND

The function hash_display() should display the following content :
T#H#E# #W#O#R#L#D# #I#S# #R#O#U#N#D#

Files (exercise)

9- Write a function display_words() in python to read lines from the text file , and display those words, which are less than 4 characters.

5- write python program to create new file “myfile.txt” and write the following lines in it:

my name is Ahmed
I have a car and bicycle

6- write Python program to write the line “ ahmed hussein” to the end of this file:

7- write Python program to read the content of one file line by line and stor it in another file line by line.

Exception handling